

Comparing the hydrology of grassed and cultivated catchments in the semi-arid Canadian prairies

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Abstract:

At the St Denis National Wildlife Area in the prairie region of southern Saskatchewan, Canada, water levels in wetlands have been monitored since 1968. In 1980 and 1983 a total of about one-third of the 4 km² area was converted from cultivation to an undisturbed cover of brome grass. A few years after this conversion all the wetlands within the area of grass dried out; they have remained dry since, whereas wetlands in adjacent cultivated lands have held water as before. Field measurements show that introduction of undisturbed grass reduces water input to the wetlands mainly through a combination of efficient snow trapping and enhanced infiltration into frozen soil. In winter, the tall brome grass traps most of the snowfall, whereas in the cultivated fields more wind transport of snow occurs, especially for short stubble and fallow fields. Single-ring infiltration tests were conducted during snowmelt, while the soil was still frozen, and again in summer. The infiltrability of the frozen soil in the grassland is high enough to absorb most or all of the snowmelt, whereas in the cultivated fields the infiltration into the frozen soil is limited and significant runoff occurs. In summer, the infiltrability increases for the cultivated fields, but the grassland retains a much higher infiltrability than the cultivated land. The development of enhanced infiltrability takes several years after the conversion from cultivation to grass, and is likely due to the gradual development of macropores, such as root holes, desiccation cracks, and animal burrows. Copyright © 2002 Crown in the right of Canada. Published by John Wiley & Sons, Ltd.

KEY WORDS wetland; prairie; hydrology; land use; infiltration; grassland

INTRODUCTION

In the semi-arid northern plains of North America there are millions of small wetlands, also called ‘prairie potholes’ or ‘sloughs’. These play an important role in the hydrology of the region and provide a critical breeding habitat for migratory waterfowl. The wetlands are vulnerable to the effects of climate change and changes of land use on the surrounding uplands (Covich *et al.*, 1997). Maintaining the hydrological and ecological functions of the wetlands is a major concern for agencies concerned with wildlife and water resources in the region.

The water balance of the prairie wetlands is governed by the cold semi-arid climate, with potential evaporation (600 to 1000 mm) exceeding precipitation (300 to 500 mm). The winters are cold and long, so that the soils freeze to depths of 1 m or more. Winter snowfall accumulates to amounts ranging from less than 10 mm up to 150 mm of water equivalent. Movement of snow-derived water from the uplands into the wetlands, as blowing snow and as runoff of snowmelt water over frozen ground, is critical to the existence of the wetlands (Labaugh *et al.*, 1998). In summer, the water balance of the wetlands is negative because evapotranspiration exceeds rainfall; surface runoff occurs only rarely, during intense rain events. The underlying glacial deposits consist largely of clay-rich glacial till with low permeability, so that groundwater

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flow is generally slow and represents a minor component of the water balance (van der Kamp and Hayashi, 1998).

Earlier studies of soil moisture and hydrology in the prairies suggest that land use on the surrounding uplands can have an important effect on wetland water balance. Christie *et al.* (1985) found that the soil beneath permanent grass was much drier than beneath adjacent cultivated land. Gray *et al.* (1985) concluded that frozen soils are capable of absorbing all snowmelt water if they are dry and have a macropore structure, whereas saturated unstructured soils have very little infiltration potential when they are frozen. In a review of grassland hydrology, de Jong and Kachanoski (1987) concluded that overwinter soil water recharge (i.e. net infiltration during the winter and early spring) on the prairies is typically 50 to 100 mm under tall stubble or brush and practically zero for fallow land. Blackburn *et al.* (1991), using the Water Erosion Prediction Project (WEPP) model, predicted that introduction of permanent brome grass cover leads to reduction of rainfall runoff. Euliss and Mushet (1996) found that, during the exceptionally wet summer of 1993, the water level rises due to rain and overland flow were smaller for wetlands in grasslands than for wetlands in cultivated lands. These studies suggest that runoff of snowmelt water and summer precipitation is likely to be greater from cultivated fields than from brush or tall grass.

At the St Denis National Wildlife Area (SDNWA), in Saskatchewan, Canada (see the site description below), water levels in about 100 wetlands have been monitored since 1968. Between 1980 and 1983 about one-third of the SDNWA was converted from dryland cultivation to permanent undisturbed grass, with the intention of providing improved nesting cover for waterfowl. As described by van der Kamp *et al.* (1999), wetlands in the grassed area dried out within a few years after the conversion to grass. Although the effect was unambiguous, the reasons for this change of hydrological behaviour were not identified.

This paper compares runoff-producing processes in cultivated fields and in the grassland at the SDNWA and identifies the causes of the drastic change of wetland water balance. The initial land use conversion was not considered as a hydrological experiment, and no hydrological data were collected before and after the conversion, except for the wetland water levels and some early data on development of the grass. The present study, therefore, relied mostly on new measurements that focus on the hydrological contrast between the cultivated land and the grassland.

STUDY SITE

The SDNWA, located in central Saskatchewan (106°06'W, 52°02'N), comprises about 385 ha and contains numerous wetlands (Figure 1). The area is characterized by moderately rolling knob-and-kettle moraine with slopes varying from 10 to 15% (Miller *et al.*, 1985) and elevation ranging between 545 and 560 m above sea level.

The area is underlain by about 100 m of glacial deposits, consisting mostly of clay-rich glacial tills with low permeability (Hayashi *et al.*, 1998a). Sands are encountered sporadically within these deposits, but the only sand layer that may have continuity across the site is a 1 to 2 m thick layer of sand at a depth of about 30 m. The soils of the SDNWA have been studied in detail by Miller *et al.* (1985). The upland soils are described as dominantly Orthic Dark Brown Chernozems [fine-loamy, mixed, frigid, Typic Haplustolls (Soil Survey Staff, 1999)] on mid slopes and lower slopes with Calcareous Dark Brown Chernozems (Entic Haplustolls) and Orthic Regosols (Typic Ustorthents) occurring on knolls, all developed from moderate- to fine-textured unsorted glacial till. Such clay-rich soils and underlying glacial deposits typify much of the northern glaciated plains.

The annual mean air temperature at Saskatoon Airport, about 40 km west of the SDNWA, is 2 °C, with monthly means of -19 °C in January and 18 °C in July (Atmospheric Environment Service, 1997). The area generally becomes covered by snow in November. Snowmelt runoff occurs in several episodes in March and April, and is occurring increasingly early due to climate warming, which is especially marked in February and March (Cutforth *et al.*, 1999; Zhang *et al.*, 2000). The 30 year mean annual precipitation in Saskatoon

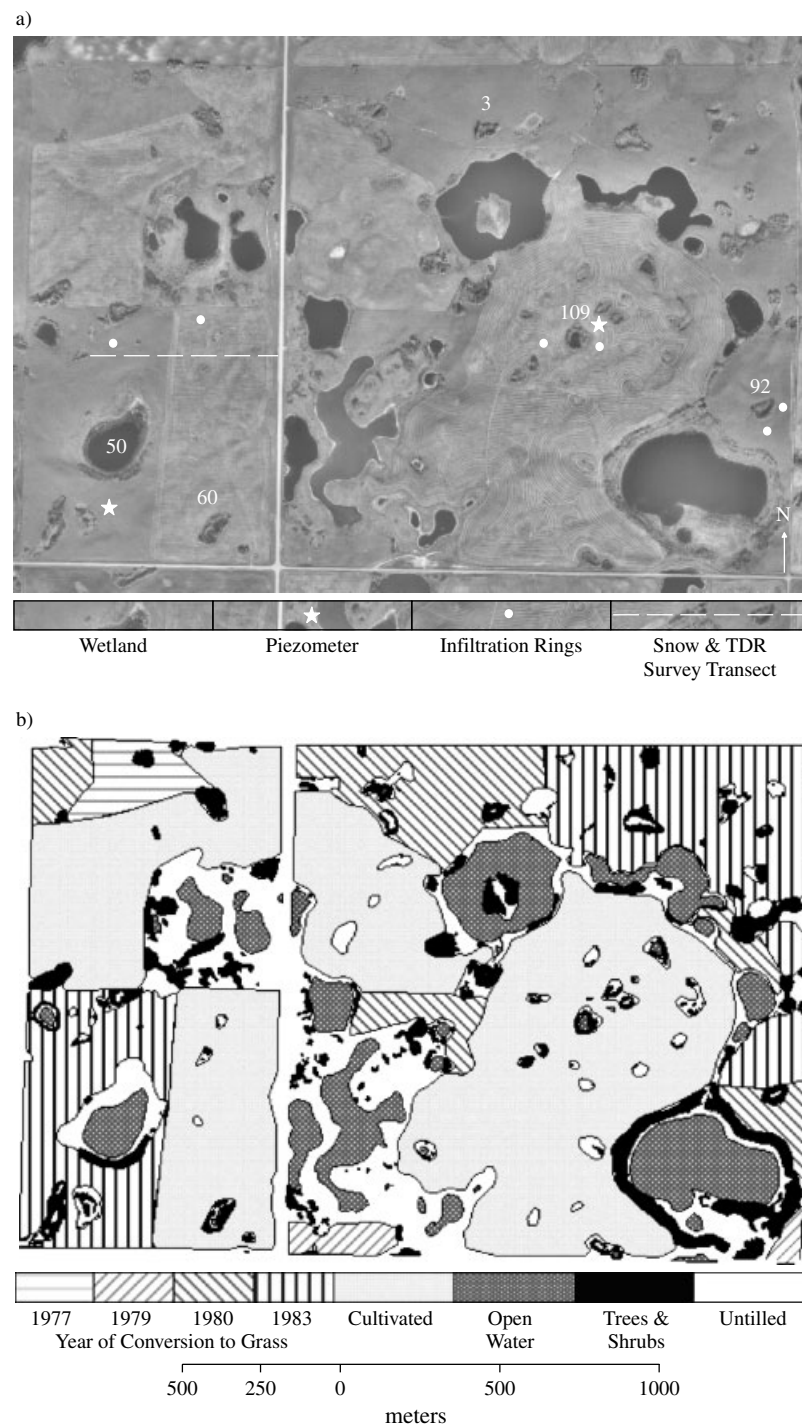


Figure 1. Plan of the SDNWA based on an aerial photograph taken in October 1997. (a) Site plan of SDNWA. Wetlands are identified by number. (b) Map of vegetation within the SDNWA boundary. The years when conversion to grass was carried out are indicated

for 1967–96 is 358 mm, of which 74 mm occurs in winter (November–April), mostly as snow. Most of the rainfall occurs in summer, May through August, during convective storms. The annual evaporation from large lakes in this area is 690–710 mm (Morton, 1983), based on the data from Last Mountain Lake, located 100 km southeast of the study site.

Aerial photographs indicate that the cultivated land within the SDNWA did not change much in extent between 1951 and 1968 and was subjected to dryland agriculture, typically a rotation of cereal crops and summer fallow in which the field was cultivated without seeding to conserve soil moisture and to control weeds. The SDNWA site became a National Wildlife Area in 1968, and water level recording was started at that time. The cultivated land remained in cultivation, with little change, between 1968 and 1980. In two stages, October, 1980 and early in 1983, a total of about one-third of the SDNWA was converted from dryland cultivation to a dense nesting cover of smooth brome grass (*Bromus inermis*) and alfalfa (*Medicago sativa*) for the purpose of improving wildlife habitat. (Small parts of the SDNWA had been previously converted in this manner.) This dense nesting cover has not been disturbed by grazing, mowing or burning. Figure 1b indicates the extent of the areas of brome grass, cultivated land, untilled land, and ‘willow rings’ around the wetlands, and the dates when the brome grass was introduced to various parts of the SDNWA. The extent of the ponds shown in Figure 1a represents conditions in October 1997, when wetland water levels within the cultivated area were higher than the normal October condition (van der Kamp *et al.*, 1999).

PREVIOUS STUDIES IN THE SDNWA SITE

Wetland water levels

At the SDNWA the water levels in ponds have been monitored by point measurements since 1968. A steel stake was driven into the ground at the deepest part of each wetland basin to serve as a reference point. Owing to frost action, the stakes were subject to considerable heaving (Shjeflo, 1972); therefore, the water levels could not be reliably measured with respect to the stakes. Instead, the depth of water in the vicinity of the central stake was measured with a graduated stick with a 4 cm diameter circular plate attached at the bottom (Conly and van der Kamp, 2001). The data are thus reported part of each pond. The water depth measurements are generally accurate and reproducible to within 5 cm.

Figure 2a shows the water level records for the years 1970 to 2000 in wetlands 60 and 109, located within cultivated fields (location: Figure 1). Water levels are not shown for the winter seasons because the water becomes largely or entirely frozen and covered with snow, so that water levels cannot be meaningfully defined or measured. As shown in Figure 2a, the typical annual cycle has maximum water levels in the spring after snowmelt, attesting to the importance of snowmelt water for the water balance of the wetlands. The notable exception occurred in 1991, when there was very little snowmelt runoff, but some of the wetlands (e.g. wetland 109, see Figure 2a) re-flooded in June due to heavy rains.

Wetlands 3 and 92 are within the areas that were converted to dense nesting cover (location: Figure 1b). Figure 2b shows that spring runoff continued to occur in these wetlands until 1986, five growing seasons after the conversion to grass for wetland 3 and three growing seasons for wetland 92. They dried out in 1987 and have remained almost entirely dry ever since, including during the high rainfall runoff event of June 1991 and the very high snowmelt runoff event of 1997. Other small wetlands within the cultivated fields and within the grassed area behaved similarly (van der Kamp *et al.*, 1999).

The inter-annual variability of peak water levels shown in Figure 2a mainly reflects climatic variability, particularly the variation of snowmelt runoff from year to year. Winter and Rosenberry (1998) described wetland water level data for North Dakota, USA, that show similar inter-annual changes. Sorenson *et al.* (1998) reported a similar pattern of fluctuation in the number of wetlands holding water in the US portion of the region, as did Conly and van der Kamp (2001) for the Canadian portion.

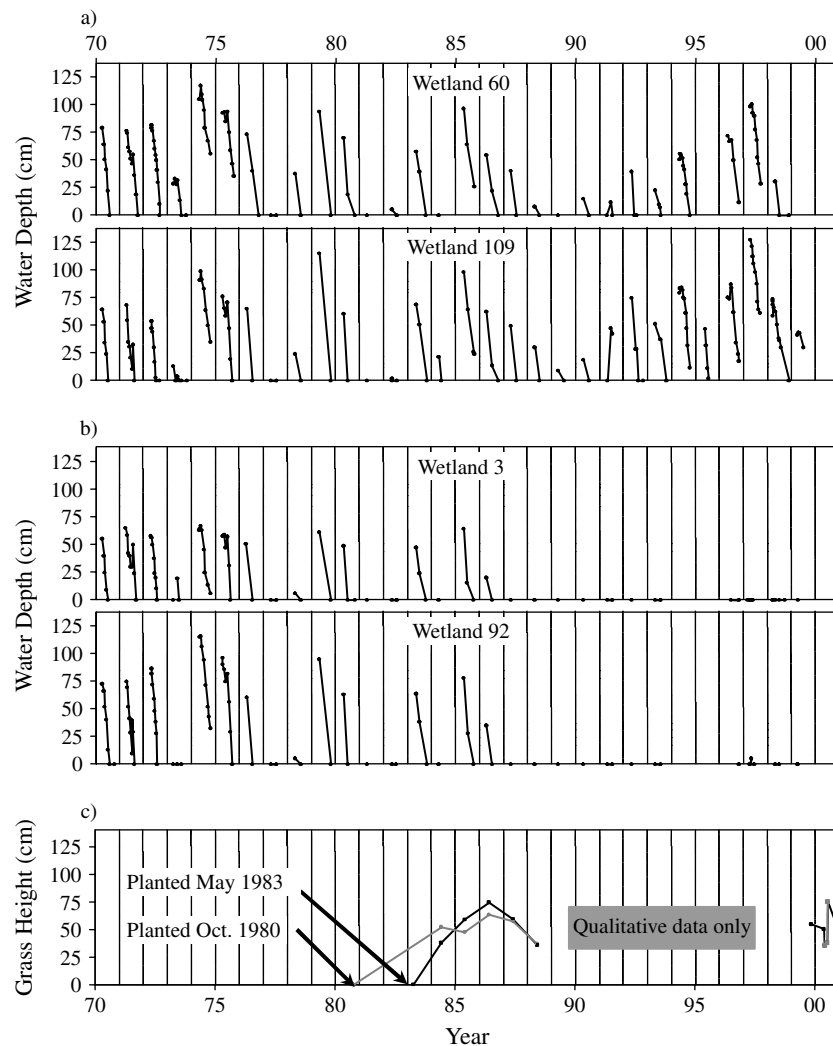


Figure 2. (a) Water depth records for wetlands 60 and 109 in the cultivated areas. (b) Water depth records for wetlands 3, and 92, in the grass areas. (c) Average maximum height of grass in the areas that were converted to grass in October 1980 and in May 1983

Water balance studies

Su *et al.* (2000) estimated the water balance of a large wetland (# 50) within the SDNWA using a lumped hydrological model for a 28 year period. They found that, of the 558 mm average annual water input to the wetland, 27% is in the form of drifted snow and snowmelt runoff from the surrounding catchment, 10% comes from summer rainfall runoff, 0.8% from groundwater seepage, and the remaining 62% from direct precipitation on the wetland. Hayashi *et al.* (1998a,b) studied the water balance of wetland 109 and found that, in 1994, 52% of the 786 mm water input to the wetland consisted of drifted snow and snowmelt runoff and 48% of the input was by summer rainfall on the wetland, with no significant runoff resulting from rainfall. They observed only one summer runoff event in the 3-5 year study period. These quantitative water balance studies confirm that snow-derived water from the catchment and rainfall directly on the wetland represent the dominant water inputs to the wetlands at the SDNWA, whereas rainfall runoff from the catchment is a minor component of the wetland water balance. Heavy rain on frozen soil during early spring could generate

runoff, but such rain events are rare and any runoff generated cannot be easily distinguished from the effects of snowmelt that would likely be occurring at the same time. Rainfall in the fall (September–November) is light and is unlikely to generate runoff because the soils tend to be dry at that time.

Development of dense nesting cover

The density and height of the dense nesting cover were monitored from 1984 to 1988 in relation to studies of duck nesting success (Clark and Shutler, 1999). The average maximum height of the vegetation was determined in mid-summer by measuring the maximum height of the vegetation at numerous 0.1 m² plots (R. G. Clark, personal communication). The data (Figure 2c) show that the brome grass planted in October 1980 was established to full height by 1984, and the grass planted in the spring of 1983 was well established by the summer of 1984 and had gained full height by the summer of 1985, when moisture conditions were excellent. The decrease of height in 1987 and 1988 is probably mostly related to the dry conditions during those years. There have been no grazing, haying, or fire in the grass areas from the time of planting until 2000, with the exception of some haying around wetland 3 in 1982.

Small mammal abundance

Studies of small mammal abundance in idle pasture, delayed hay, and dense nesting cover, including dense nesting cover at the SDNWA (Pasitschinnak-Arts and Messier, 1998), showed that the dense nesting cover had the highest abundance of small mammals. Casual observation indicates an abundance of small mammal burrows and 'mole heaps' in the dense nesting cover at the SDNWA, whereas there is very little activity of this kind in the cultivated fields. Since small burrows could act as efficient water infiltration galleries, they may have a bearing on the hydrological characteristics of the undisturbed dense cover.

STUDY METHODS

Vegetative cover and land use

Records of crop types and of stubble height going into winter have been kept for the various cultivated fields at the SDNWA since 1993. Stubble heights were estimated by simple measurements using a metre stick at a number of locations within each field. The height of the dense nesting cover was estimated each fall and spring since 1998, with notes as to overall distribution and appearance. The condition of the brome grass between 1988 and 1998 was qualitatively estimated from photographs and visual observations made during snow surveys at the end of the winter. Since 1968, notes were kept on the conditions of the vegetation, including trees in and around the wetlands, and these were checked by means of aerial photographs that were taken in 1968, in 1997, and at several other times.

Soil moisture

In the fall of 1999 and 2000 the soil moisture was measured along a west–east line north of wetland 50 through a section of brome grass and into a cultivated field (location: Figure 1a) and along several transects across wetlands 109 and 92 (not shown in Figure 1a). Along each transect the soil moisture was measured to a depth of 20 cm with time domain reflectometry (TDR) (Soilmoisture Equipment Corp., Trase System). Soil core samples (0–20 cm) were taken at selected locations along these transects where TDR measurements were obtained and the volumetric water content was determined by the oven-drying method, in which the samples were weighed before and after drying at 105 °C. These determinations served to verify and calibrate the TDR measurements. Core samples to a depth of 0.9 m were also collected using a truck-mounted auger at ten locations in the cultivated field near wetland 109 (29–30 September 2000) and at nine locations in the brome grass near wetland 50 (17–18 October 2000), and the porosity and volumetric water content were

determined by the oven-drying method. There was no significant precipitation in the time interval between the two dates of coring.

Snow surveys

Snow surveys were conducted during March for years 1994 to 2000 (1995 excepted) along several transects, including those where soil moisture was measured in 1999 and 2000. The surveys were timed to catch the maximum snow accumulation before snowmelt. Vegetation types, such as willow rings, wetland vegetation, brush, and cultivated fields (including crop type and stubble height), were noted. Snow depth was recorded at either 10 or 20 m intervals by inserting a 6 cm diameter snow collection tube into the snowpack. At several locations representative of each of the various vegetation types, the contents of the tube were collected and later weighed for snow-water equivalent (SWE) determination and calculation of snow density. Average snow densities were determined as a function of snow depth and vegetation type and the snow depth measurements were converted to SWE using the appropriate density for each year and vegetation type.

Surface runoff

Numerous small closed depressions in the landscape of the SDNWA provide a convenient method to estimate snowmelt runoff generated within their respective watersheds. At the time of snowmelt the ground below the depressions is still frozen and the runoff water remains trapped in the depressions for up to 10 days before it begins to infiltrate. Assuming that the peak water level in a depression represents all of the snowmelt water that has entered the depression, one can estimate the total runoff. However, a small but unknown portion of water infiltrates into the soil below the depressions before the peak water level is reached, so the total runoff from the watershed is slightly underestimated by this method. Three depressions were selected in the cultivated field near wetland 109, where soil moisture and snow accumulation were measured. Water levels in the depressions were recorded and the volume of water in each depression was estimated using a volume–depth function (Hayashi and van der Kamp, 2000) based on detailed topographical maps. Small depressions in the grassed area were also inspected regularly during the snowmelt season for any sign of runoff and water accumulation.

Infiltrability

The infiltrability of the soils in the cultivated fields (cultivated soil) and in the planted dense cover (grassed soil) were measured with single-ring infiltrometers. In July and October 1998 and in October and November 1999, metal rings of 20–25 cm in height and 30–40 cm in diameter were hammered into the soil until the top of the ring protruded 3 to 5 cm above the soil surface. Seventeen such rings were installed in the cultivated fields and 15 in the dense cover. Near each set of two or three rings a set of thermocouples was installed at depths of 0, 20, 40, 60, 80, 100, and 150 cm.

The infiltrability of the soil within each ring was measured in March 2000 during snowmelt while the soil was still frozen at 20 cm depth and deeper, as indicated by the sub-freezing temperature of nearby thermocouples. A known volume of ice-water was poured into each ring, and the subsequent change of water level was measured while maintaining a reasonably constant head in the ring by periodically refilling to the original water level (constant-head test). The falling-head method was used if the rate of infiltration was so slow that only a few millimetres of water level change were recorded over several hours. The infiltration tests were repeated on the same infiltration rings in April and August 2000, when the ground was entirely thawed.

Piezometer construction and groundwater level monitoring

Piezometers and water table wells were installed in 1980 within and around wetland 109 (Miller *et al.*, 1985), in the brome grass area around wetland 50 in 1989 (Woo and Rowsell, 1993) and in the middle of wetland 92 in the fall of 1999. They were made of PVC pipes and installed in augered holes, sealed

by bentonite pellets to the surface. The locations of the piezometers referred to in this paper are shown in Figure 1.

Water levels have been measured in the piezometers within and near wetland 109 from 1980 to 1985 and from 1994 to the present; in the piezometers near wetland 50 since installation in 1989, but with only sporadic measurements from 1995 to 1997; and in the piezometer in wetland 92 since installation. Slug tests (e.g. van der Kamp, 2001) were carried out on all the piezometers to check that they were responding quickly enough to follow seasonal changes of piezometric level.

RESULTS

Vegetative cover and land use

During the period 1993–2001 the cultivated field around wetland 109 (location: Figure 1a) was in summer fallow only in 1995. Stubble height going into winter has ranged from 5 to 45 cm, depending on the crop type. In most years the stubble height exceeded 25 cm. A similar cultivation pattern was noted for the other cultivated fields. Data on stubble heights during the 1970s and 1980s are not available, but it is likely that stubble heights were smaller, reflecting the size and type of farm machinery. The height of the brome grass has generally been in the 40 to 80 cm range (Figure 2c).

Changes of the trees in the ‘willow rings’ around the wetlands have been documented from year to year during the period 1968–2001. Generally, there has been slow expansion of the trees. One particular case of interest is wetland 109, where there were no trees at all in 1968, but which had a dense tree ring by 2001, as attested by observation (J. B. Millar, personal communication) and by aerial photographs. By contrast, there has been little change in the tree ring around wetland 60. This observation is noteworthy because the water level data for these two wetlands (Figure 2) suggest that the development of trees around wetland 109 did not result in a significant difference between the water balances of the two wetlands.

Porosity and soil moisture

The soil water data for the transect north of wetland 50 (location: Figure 1) show that the soil water content of the top soil (0–0.2 m) was generally lower under the grassed uplands than the cultivated uplands (Figure 3). The difference was especially marked in 2000, when the cultivated field was fallow during the preceding summer. Data from other transects in the vicinity of wetlands 109 and 92 show similar results.

Table I summarizes the average porosity and moisture content in the fall of 2000 for ten test sites in the cultivated field near wetland 109 and for nine sites in the grass around wetland 50. A crop of field peas had been grown in the field around wetland 109 during the summer of 2000. These results show that the cultivated soil has a higher water content not only in the top 0.2 m (Figure 3), but also to a depth of 0.9 m; this is consistent with the findings of other workers (e.g. Christie *et al.*, 1985), and indicates that in the semi-arid prairie region a permanent grass cover leads to lower soil moisture. Air-filled porosity, an important parameter

Table I. Volumetric soil moisture contents, total porosity, and air-filled porosity for soil samples from cultivated and grassed fields, September and October 2000

Depth (cm)	Volumetric soil moisture (%)		Total porosity (%)		Air-filled porosity (%)	
	Cultivated	Grass	Cultivated	Grass	Cultivated	Grass
0–15	16.3	11.8	54.5	61.8	38.2	50.0
15–30	21.0	14.3	47.4	50.2	26.4	35.9
30–60	17.3	12.3	47.8	45.5	30.4	33.2
60–90	22.5	19.8	42.7	42.1	20.2	22.2

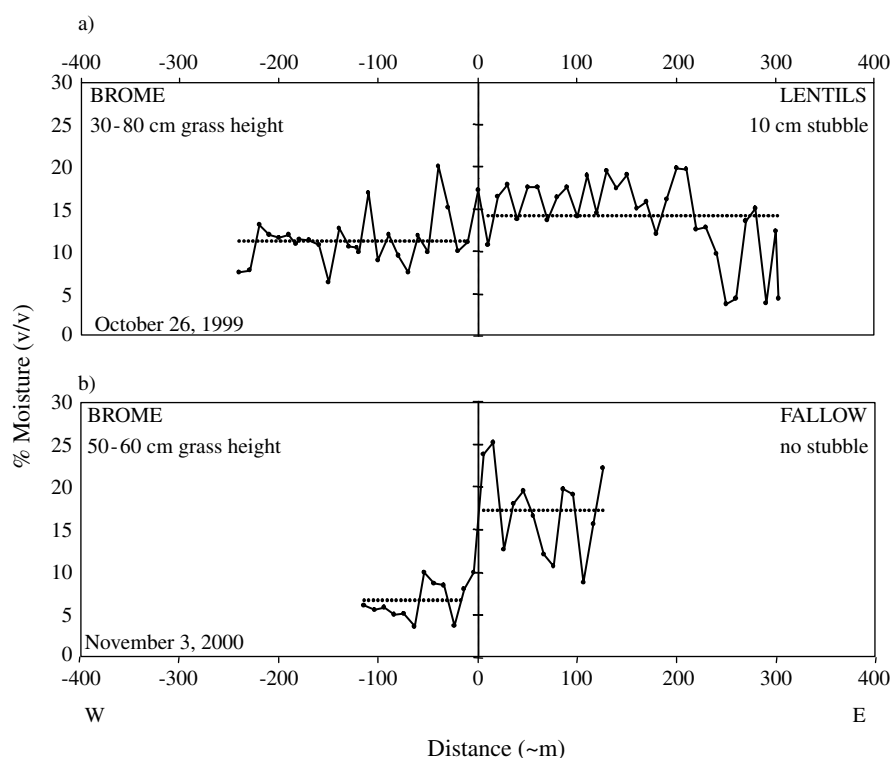


Figure 3. Volumetric soil moisture (0 to 20 cm) along a west-east transect from grass to cultivated land north of wetland 50 in the SDNWA, as determined by TDR and soil cores in (a) the fall of 1999 and (b) the fall of 2000

for frozen-ground infiltration (Gray *et al.*, 1985; Seyfried and Murdock, 1997), is higher in the grassed area to a depth of at least 0.3 m, through the combination of higher total porosity and lower water content.

Snow distribution

The distributions of snow in the years 1997 to 1999 are shown in Figure 4 for the transect north of wetland 50 (location: Figure 1). The type and height of the vegetative cover going in to winter is noted in Figure 4. In 1998, the snow in the grass had already partially melted by the date of the snow survey, as indicated in Figure 4 by the points where there was little or no snow remaining.

The snow survey data indicate that: (a) summer fallow leads to a high variability of snow cover; this is thick in the wetland margins and along the plot boundaries, and thin in exposed areas such as hill tops and smooth terrain; (b) there is little difference in patterns of snow accumulation between the brome grass areas and cultivated fields with tall stubble.

The average snow density for all measurements taken at the SDNWA during the years 1996–2000 in the brome grass and in the cultivated fields with tall stubble was practically the same (0.21 g cm^{-3}), but the density in the fallow fields was higher, averaging 0.28 g cm^{-3} . A slight tendency for higher densities was noted when snow depths exceeded about 60 cm, as mentioned by Pomeroy and Gray (1995). For the present study, snow depths rarely exceeded 60 cm in the grass or cultivated fields and, therefore, a constant density, independent of snow depth, has been used. During the snow surveys it was noted that the snow in the grass areas melted earlier than the snow in the cultivated fields with tall stubble (Figure 4b). Within the grass areas there is considerable vegetation protruding above the snow in most years, as indicated by the dark appearance

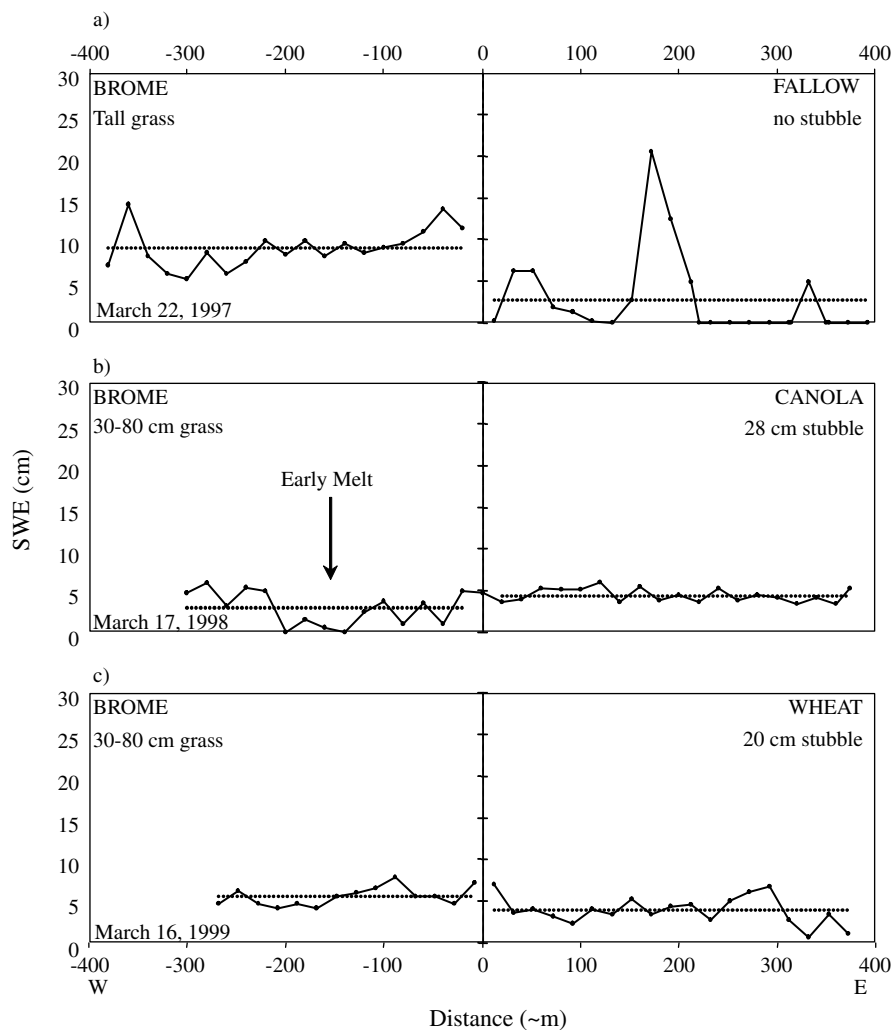


Figure 4. SWE distribution at the end of the winter for the years (a) 1997, (b) 1998 and (c) 1999. Transect runs west–east through brome grass and cultivated land north of wetland 50

of these areas when seen from a distance. Presumably, the resultant lowering of the albedo leads to earlier snowmelt (Pomeroy *et al.*, 1998).

Snowmelt runoff

Table II summarizes the snowmelt runoff into three small depressions (109SE, 109S and 109SW) on cultivated fields near wetland 109 in the spring of 2000 and 2001, SWE before snowmelt in their catchments, and soil water content in the preceding fall. The fields had wheat stubble with an average height of 30 to 45 cm during the winter of 1999–2000, and pea stubble with a height of 5 to 10 cm during the winter of 2000–01. The estimated runoff was smaller than the available SWE, indicating that some snowmelt water infiltrated into the frozen soil. The relative proportions of infiltration and runoff are variable from year to year and place to place within the cultivated fields.

In contrast to the cultivated fields, no visible signs of snowmelt runoff were encountered in the grass areas during frequent site visits in 1999 and 2000, either as flowing water or as puddles in depressions, with the

Table II. Catchment area, topsoil water content, SWE, and snowmelt runoff in the grassed area and for three depressions in the cultivated field

Depression	Catchment area (m ²)	Soil moisture (%)		SWE (mm)		Measured snowmelt runoff (mm)		Theoretical snowmelt runoff (mm)	
		20–25 Oct 1999	1–3 Nov 2000	26 Feb 2000	2–3 March 2001	March 2000	March 2001	March 2000	March 2001
Grass	–	11	6.7	>25	49	0	0	?	6
109SW	2030	19	15	29	39	22	3.4	7	9
109S	6920	24	18	35	37	13	2.6	14	10
109SE	2910	25	21	33	44	19	3.1	13	17

exception of small puddles along a track where motor vehicles had passed repeatedly during the previous summer.

Infiltrability

The single-ring infiltration tests for frozen and unfrozen soils are designed to serve as a comparative indicator of the infiltration rate under different conditions: grassed and cultivated, and frozen and unfrozen. However, quantitative interpretation of infiltration data is made difficult by a number of factors, including soil heterogeneity, macropore flow, and phase change between ice and liquid water, that are not included in the theory of infiltration tests (e.g. Reynolds and Elrick, 1990). A similar approach was adopted by Kane (1980), who used auger-hole infiltration rate to compare the infiltrability of a silt loam soil under frozen and

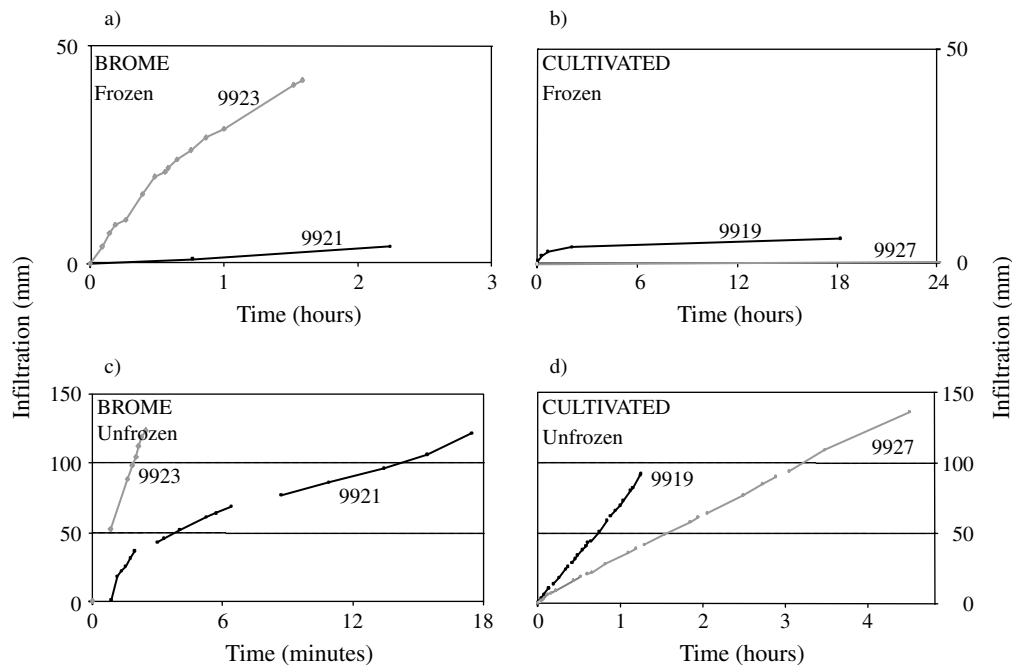


Figure 5. Measured infiltration as a function of time for individual tests carried out in grass (rings 9921 and 9923) and cultivated land (rings 9919 and 9927). Infiltration tests shown in (a) and (b) were performed in frozen conditions during the snowmelt season of 2000; tests (c) and (d) were performed in unfrozen conditions during the summer of 2000. Note the differing time scales

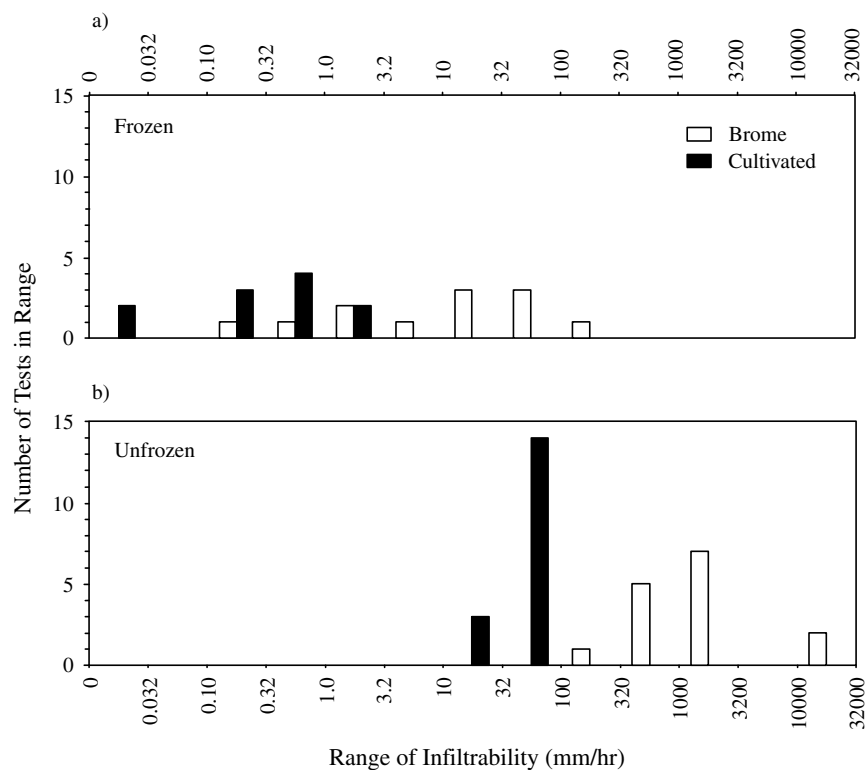


Figure 6. Range of infiltrability in cultivated versus grassed fields for single-ring tests in (a) frozen and (b) unfrozen conditions

unfrozen conditions. Figure 5 shows examples of time–infiltration plots for cultivated soil near wetland 109 and grassed soil near wetland 92.

There was a marked difference in the infiltrability between the grassed and the cultivated soil, especially while the soil was frozen (Figure 6). Under frozen conditions, the infiltrability of the cultivated soil was very small, whereas the infiltrability of grassed soil was moderately high and showed a large variation (Figure 6a). Under unfrozen conditions the infiltrability of the cultivated soil was much higher than when it was frozen, but still significantly lower than that of the grassed soil (Figure 6b). The frozen-soil tests were carried out during the latter portion of the snowmelt period, when the snowpack had been partially ablated. As indicated by the snowmelt runoff results for small depressions, it is likely that some snowmelt water infiltrated into the soil prior to the tests and froze again, thereby reducing infiltrability (Zhao *et al.*, 1997), at least in the cultivated fields.

Groundwater levels

Figure 7 shows the water levels in a water table well (80W12) northeast of wetland 109, located on a knoll within the cultivated field, and in a piezometer (S3-12) located in the brome grass on the upland south of wetland 50. The screen of S3-12 is located 3.6 m below the ground surface, which is 6 m above the bottom of wetland 50. The water levels in both 80W12 and S3-12 experienced sharp rises in the spring time, presumably in response to the recharge of groundwater by snowmelt water. Other shallow piezometers in the cultivated fields and in the brome grass showed similar responses to snowmelt. Unfortunately, no groundwater data are available for the areas converted to brome grass prior to the conversion, and thus the available data provide no clear indications as to whether the introduction of permanent grass leads to significant changes in the elevation of the water table.

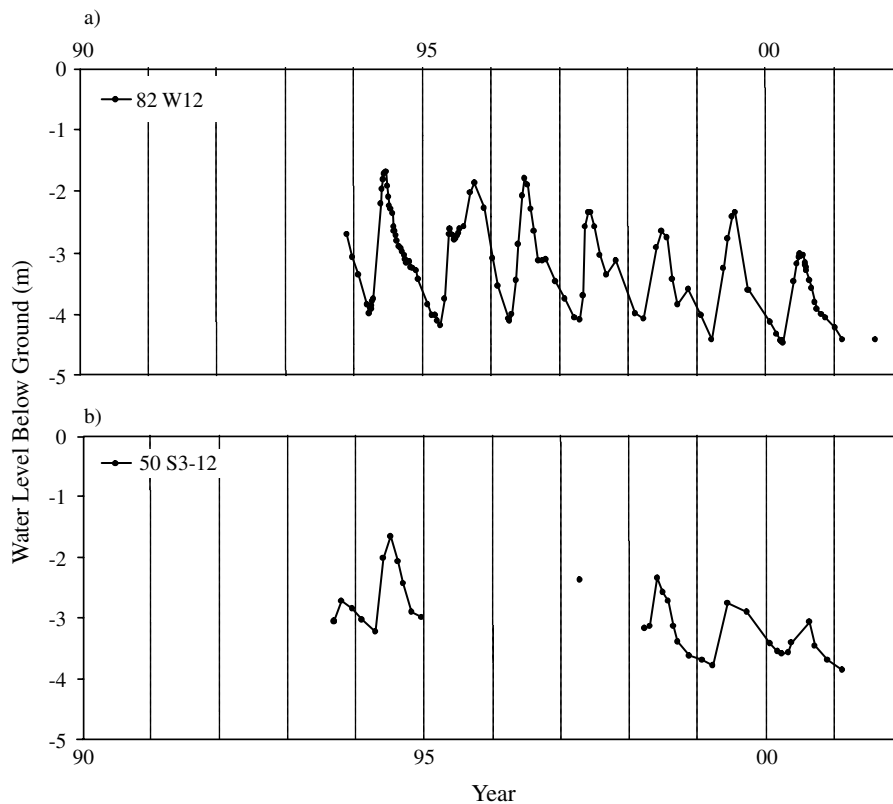


Figure 7. Groundwater levels in (a) a water-table well (80W12, 5.6 m) in the cultivated land northeast of wetland 109 and (b) shallow piezometer (50S3-12, 3.6 m) in the grassland south of wetland 50

The water table under wetland 92 fluctuated between 7.1 and 7.2 m below the ground surface in 1999–2001. This is in a marked contrast to wetland 109, in which the water table came up near or above the ground surface every spring, and stayed within 2–3 m of the ground surface even during the driest period. Very likely the water table under wetland 92 was also near or above ground surface in the years prior to 1987, when there was generally standing water in the wetland throughout most or all of the summer (Figure 2). The 7 m decline of the water table under wetland 92 between 1987 and 1999 indicates that a moisture deficit has accumulated in the wetland soil over the years after the wetland dried out.

DISCUSSION

The drying out of the wetlands in the permanent grassland implies that evapotranspiration output from these wetlands, including the uptake by marginal vegetation, now exceeds the input of water by direct precipitation, by wind transport of snow, and by surface runoff of snowmelt water and of rain water. The surface runoff of rain water is a minor component of the water balance even in the cultivated fields, and is even smaller in the grasslands, as indicated by the enhanced infiltrability of the grassed soils (Figure 6). Thus, the drying effect of the grass can occur only if transport of snow-derived water into the wetlands is greatly reduced. This implies that: (a) snow must be trapped on the upland so that wind transport of snow into the wetlands is greatly reduced; (b) snowmelt water derived from the snow on the upland must infiltrate rather than running

off into the wetlands, and must then be lost by evapotranspiration from the uplands. The field measurements provide information on how these two mechanisms operate in the grasslands compared with the cultivated fields.

Figure 4 shows that permanent grasslands are more capable of trapping snow and keeping it on the upland than fallow fields. However, tall stubble in the cultivated fields can trap just as much snow as permanent grass. Therefore, snow-trapping by tall permanent grass itself is not sufficient to lead to drying out of wetlands, although it may contribute by reducing input of windblown snow. The height of the brome grass is generally in the range of 0.4 to 0.8 m, whereas the snow depth averages about 0.35 m by the end of the winter. Visual inspection of snow conditions and snow surveys indicate that significant snow drifting does occur in the brome grass in winters when the total snowfall is above average and if there have been high winds. In such years there can certainly be significant input of water into the wetlands by wind transport of snow. However, this water input is apparently now insufficient to overcome the accumulated soil moisture deficit in the dried out wetlands (e.g. wetland 92), and ponding of surface water in the wetlands is at most minor and short-lived (Figure 2b). In summary, our field data indicate that the change in snow-trapping efficiency is a necessary but not sufficient cause of drying out of the wetlands in the grassed watershed.

Infiltration of snowmelt water into frozen ground is the critical process that controls the amount of snowmelt runoff. Single-ring infiltrometer tests clearly indicated that the infiltrability of the frozen soil is much higher in the grass than in the cultivated field (Figure 6). For most of the infiltration test locations, the frozen soil in the grassed areas has infiltrability in excess of the typical rate of snowmelt, which is unlikely to exceed 10 mm hr^{-1} (Woo and Rowsell, 1993). By contrast, all the infiltration tests on frozen soil in the cultivated fields indicated an infiltrability considerably less than 10 mm h^{-1} . These test results, therefore, imply that the grass areas should produce little or no snowmelt runoff, whereas the cultivated fields will generate significant snowmelt runoff.

Gray *et al.* (1985) analysed the snowmelt infiltration data from many sites in Saskatchewan and proposed a simple equation for estimating the total infiltration in frozen soils with 'limited' infiltrability; in other words, soils without macropores or cracks. Their equation is written as

$$\text{INF} = 5(1 - \theta_p)\text{SWE}^{0.584} \quad (1)$$

where INF (mm) is the amount of total infiltration, θ_p is the pre-melt soil water content in the 0–30 cm zone expressed as the degree of saturation, and SWE (mm) is the available snow-water equivalent.

To assess the sensitivity of runoff to soil dryness, Equation (1) was used to estimate the amount of runoff (= SWE – INF). The average fall soil water content of the 0–20 cm depth range (Table II) was used in the calculation assuming that it represents pre-melt water content, which was not measured in this study. The average porosity of the 0–30 cm depth range (Table I) was used to convert volumetric water content to the degree of saturation.

The results of the calculations are summarized in Table II for the transect in the grass north of wetland 50 and for the three small depressions in cultivated land near wetland 109. Equation (1) predicts measurable amounts (6–17 mm) of runoff in both cultivated fields and grass areas. The predictions are roughly consistent with the measured runoff in the cultivated land, suggesting that the soils in the cultivated fields at the SDNWA fall within the class of prairie soils with 'limited' infiltrability to which Equation (1) applies. In contrast, the prediction is not even qualitatively consistent with the data for the grass area, where no runoff was observed. This result indicates that the drier condition of the soil in the grass area cannot by itself account for the large difference in infiltration between the grassed and the cultivated soil (Figure 6) and the drastic reduction in snowmelt runoff. Therefore, it may be concluded that, in Gray *et al.*'s (1985) terminology, the grassed soil has 'unlimited' infiltrability, which refers to soils capable of infiltrating all snowmelt water through macropores and cracks (Johnson and Lundin, 1991).

The large difference in frozen and unfrozen infiltrability between the grassed and cultivated soils suggests that the grassed soil has a well-developed macropore network, because both soils have similar texture

characterized by high clay content. This suggestion is borne out by the higher porosity of the soil in the grassed areas (Table I). The only place in the grass where puddles were observed during snowmelt was along tracks where vehicles had repeatedly passed, indicating that the high infiltrability was reduced by compaction.

The comparison of grass height data and the water level data (Figure 2) provides additional insight into the processes that led to the decrease of snowmelt input to the wetlands. The data show that the major change in the water balance of the wetlands in brome-grass catchments took place several years after full development of the grass cover. In particular, the whole drainage basin of wetland 3 was converted to grass in the fall of 1980 (Figure 1b), and the grass was well established by 1984, yet the wetland continued to receive snowmelt runoff water until 1987. These data confirm that snow trapping on the upland by itself is not the main factor controlling the amount of snowmelt runoff, because the grass was certainly acting as an efficient snow trap by 1985, yet the wetlands continued to receive strong input of snowmelt water. Also, the data indicate that the grass cover around wetland 3 was growing vigorously by 1984 (3 years after conversion), implying strong transpiration and dry soil, yet the reduction of snowmelt runoff was not observed until the spring of 1987. This observation confirms the conclusion that soil dryness under the grassed upland is not a major factor controlling the amount of snowmelt runoff. The amount of runoff is most likely controlled by the presence of a macropore network, and the delayed hydrological response to the conversion to grass indicates that the macropores required several years to develop. The nature of the macropores has not been determined, but they could be in the form of root channels, worm holes, animal burrows, or desiccation cracks.

Elliott and Efetha (1999) reported that after 10 years of zero-till the summertime infiltrability of a prairie soil was somewhat increased. Similarly, Elliott *et al.* (2001) found that water levels in small prairie wetlands within zero-till fields were decreasing compared with the water levels of wetlands in an adjacent conventional tillage field. These findings are consistent with the findings reported in this paper, indicating that if cropped soils are not cultivated then runoff decreases.

There has been a gradual trend within the study area towards reduced cultivation, less summer fallow, and taller stubble. These changes would tend to lead to reduced runoff rather than more runoff, and hence to relatively drier conditions for the wetlands in the cultivated fields—the opposite to what was observed. Therefore, the marked contrast between the wetlands in the cultivated fields and those in the grass areas must be caused by the change in the hydrological characteristics of the grassed area, not of the cultivated fields.

The presence or absence of trees apparently does not greatly impact the water balance of the wetlands, as indicated by the similar water level regimes for wetlands 60 and 109 (Figure 2), which show no obvious evidence of the development of a tree ring around wetland 109 during the period of record. Hence, gradual changes in the area of the tree rings around the wetlands, either in the cultivated lands or the grass areas, cannot account for the abrupt change of wetland hydrology in the grass areas.

The groundwater data suggest that groundwater recharge continues to occur under the grass, but do not allow conclusions as to whether the rate of recharge increased or decreased after the introduction of grass. Groundwater recharge is equivalent to net infiltration (infiltration minus evapotranspiration). The grass areas have less runoff and higher infiltration, but also have a higher evapotranspiration, reflecting a longer period of active growth and transpiration (e.g. Blackburn *et al.*, 1991). Therefore, groundwater recharge could be either higher or lower in the grass areas compared with the cultivated fields. The distribution of the recharge is probably changed by the introduction of grass. Cultivation tends to enhance depression-focused recharge of snowmelt water (Hayashi *et al.*, in press), whereas recharge flux is more evenly distributed in the grass areas, where most snowmelt water infiltrates without running off. The observed fluctuations of the groundwater levels may also be influenced by water movement to and from the frost zone during freezing and thawing. Conclusive analysis of the observed water table changes is therefore not feasible without much more detailed study.

CONCLUSIONS

Converting cultivated fields to permanent, undisturbed grasslands reduced the wind-driven snow transport and snowmelt runoff, leading to drying out of the wetlands in the converted grass areas. Two factors operating together led to this hydrological effect: (1) tall permanent grass cover is effective in trapping snow, so that the wind-driven transport of snow into wetlands is reduced in grassed catchments; (2) the undisturbed grass cover leads to the development of a macropore network in the topsoil, which increases soil infiltrability, especially during the snowmelt period when the soil is frozen. The macropore network takes several years to develop after introduction of the grass, as indicated by the delayed hydrological response of the wetlands to the cultivated-to-grass conversion. Increased dryness of the soil beneath the grass is also a contributing factor, but its contribution is relatively minor. The net effect of the permanent cover of undisturbed tall grass is to trap snow effectively and infiltrate snowmelt water and summertime rain into the soil, where most of it is used to supply transpiration by the deep-rooted grass.

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